



Department of Computer Science & Information Technology



Course Information

Course Code: CS-08519

Credit Hours: 4 (3+1)

Lab Hours/Week: 3

Course Title: **Computer Networks**

Lecture Hours/Week: 3

Pre-Requisites: **None**

h. Detail of Technology involved_ Software Used_ Lab_ Computer Networks:

1. Cisco Packet Tracer (Version 8.2 or above).

Cisco Packet Tracer is a powerful network simulation tool developed by Cisco Systems, widely used for learning, teaching, and practicing networking concepts. It provides a virtual environment where users can design, configure, and troubleshoot computer networks without needing physical hardware. With its drag-and-drop interface, students and professionals can build complex network topologies that include routers, switches, servers, and various end devices. Packet Tracer supports a wide range of Cisco commands, enabling users to gain hands-on experience with real-world networking scenarios. It is commonly used in Cisco Networking Academy courses to develop practical skills in areas such as IP addressing, routing, switching, VLAN configuration, and network troubleshooting. Overall, Cisco Packet Tracer serves as an essential educational simulator that bridges the gap between theoretical learning and real-world network implementation.

2. Wireshark Network Analyzer:

Wireshark Network Analyzer is a widely used open-source tool designed for capturing and analyzing network traffic in real time. It allows users to inspect the data packets transmitted across a network, providing detailed insights into communication protocols, source and destination addresses, and packet contents. Wireshark supports hundreds of protocols, making it an essential tool for network administrators, cybersecurity professionals, and students studying computer networks. Its graphical interface makes it easy to visualize and filter network data, helping in troubleshooting network issues, detecting security threats, and verifying protocol implementations. By using Wireshark, users can diagnose network performance problems, identify unauthorized access, and gain a deep understanding of how data flows within a network. Overall, Wireshark is an indispensable tool for both learning and professional network analysis.